

Jagannath Sai Kasarabada

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SUMMARY

AI Data Engineer with 3 years of experience building scalable data platforms, production-grade ETL/ELT pipelines, and cloud-based data solutions. Proven expertise in data engineering, data migration, data modeling, and pipeline optimization across Azure, AWS, and GCP. Strong proficiency in Python, SQL, Snowflake, Databricks, Azure Synapse, BigQuery, AWS Glue, and Azure Data Factory, with a focus on reliable and efficient data infrastructure.

TECHNICAL SKILLS

- **Data & ML Engineering:** PySpark, Spark, Feature Engineering, Batch & Streaming Pipelines, Kafka, Structured/Unstructured Data Processing.
- **AI / GenAI:** RAG, Prompt Engineering, Embeddings, Vector Search, LLM Evaluation, Context Engineering.
- **Cloud Platforms:** Azure (Synapse, ADF, ADLS, Logic Apps), AWS (Glue, S3, DMS, SageMaker), Snowflake.
- **Distributed Systems:** Query Plan Analysis, Shuffle Tuning, Salting Techniques, Broadcast Joins.
- **APIs & Systems:** REST APIs, Microservices, Real-time Inference Pipelines, Event-driven Systems.
- **DevOps & MLOps:** CI/CD (Azure DevOps, GitHub Actions), DVC, Docker, Model Deployment, Pipeline Automation.
- **Data Governance & Security:** RBAC, Data Masking, PII Handling, Secure AI Workflows.

WORK EXPERIENCE

Data Engineer, Marathon Petroleum Corp

June 2025–Present | San Antonio, TX

- Built and maintained **production-grade data pipelines** and AI-ready feature layers supporting analytics and ML workloads at scale (1TB+ daily).
- Designed **real-time alerting and automated recovery workflows** using Azure Logic Apps, achieving 99.9% pipeline availability.
- Optimized distributed Spark workloads by 35% through partition skew handling (salting) and broadcast joins, improving latency and cost efficiency.
- Developed **CI/CD automation** pipelines using Python and PowerShell, reducing deployment time by 90% and enabling reproducible data workflows.
- Implemented **secure data access patterns (RBAC, dynamic masking)** ensuring compliance and protecting sensitive datasets used in AI systems.

Data Engineer, Fiserv

Feb 2025 - June 2025 | Atlanta, GA

- Architected a **scalable data lake (Bronze/Silver/Gold)** enabling structured and unstructured data processing for ML and analytics.
- Built **feature engineering pipelines and automated retraining** workflows using SageMaker and Snowflake, improving model inference throughput by 15%.
- Led **cloud migration from DB2 to Snowflake (AWS) using DMS and S3**, ensuring 99.9% data accuracy via validation frameworks.
- Developed reusable PySpark frameworks improving developer productivity by 60% and accelerating pipeline delivery.
- Enabled **low-latency data access** patterns supporting near real-time analytics and predictive ML use cases.

Data Analyst, Rang Technologies Inc

Feb 2024 - Feb 2025 | Atlanta, GA

- Built and optimized clinical data processing pipelines using SAS and Python, performing data ingestion, cleansing, transformation, validation, and preparation of structured datasets for downstream analytics and reporting.
- Worked with CDISC standards, including SDTM and ADaM, to design and transform clinical datasets into standardized, analysis-ready data structures while maintaining data quality, consistency, and traceability.
- Applied SQL/Python-based data analysis, biostatistics, and machine learning techniques to large clinical datasets, developing analytical workflows and predictive models to identify patterns and generate data-driven insights for clinical studies..

Engineering Intern, Changing the Present

Dec 2023 - Feb 2024 | Sunrise, FL

- Developed optimized **multi-stage ETL pipelines** using **Pandas and NumPy** to ingest high-dimensional datasets from REST APIs, reducing data preparation latency for ML training by **40%**.
- Collaborated with cross-functional teams by managing **Agile sprints** in Asana and implementing **GitHub-based CI/CD pipelines**, significantly improving project communication and delivery timelines.
- Designed and implemented **data models** and structured data schemas for high-dimensional datasets, optimizing data organization and transformations to support efficient machine learning workflows and downstream analytics.

Junior Software Engineer, Shiash Info Solutions

Nov 2021 - June 2022 | Chennai, India

- Built an **E-Auction application** that enabled merchants to list and sell products, customers to purchase products, auctioneers to conduct auctions, and administrators to manage auction operations, making the buying and selling process more efficient and reliable.
- Worked with MySQL databases, REST APIs, HTML, CSS, JavaScript, and Bootstrap to develop responsive application interfaces, integrate backend services, manage application data, and collaborate with senior developers throughout the software development lifecycle.

PROJECTS

CrediVerify:

- **Tech Stack:** Python, Streamlit, Gemini, GPT-4, PubMed Integration.
- **Objective:** Develop an automated fact-checking application to validate health claims in video content against peer-reviewed medical literature using **RAG**.

- **Outcome:** Engineered a **RAG-based pipeline** that reduced manual fact-checking time by **~80%** and achieved a **90%+ match accuracy** between video transcriptions and PubMed medical literature using Vector embeddings.

Health Whisperer:

- **Tech Stack:** Streamlit, Supabase, Telegram Bot API, OpenAI Models.
- **Objective:** Create a cross-platform wellness ecosystem unifying physical and mental health data.
- **Outcome:** Deployed an interactive chat interface that generates personalized, empathetic behavioral nudges, enhancing user engagement with wellness goals.

CERTIFICATES

- **Applied Generative AI Specialization** – Purdue University
- **Microsoft Azure Data Fundamentals**
- **SAS Certified Specialist:** Base Programming Using SAS 9.4
- **Artificial Intelligence Engineer** – Simplilearn
- **Advanced Deep Learning using Keras & TensorFlow** – IBM

EDUCATION

- **Executive PhD in Artificial Intelligence** | University of Cumberlands, KY | Aug 2026 – June 2029
- **Master of Engineering, Artificial Intelligence** | University of Cincinnati, OH | GPA: 4.0/4.0 | Aug 2022 - Dec 2023
- **Bachelor of Technology, Computer Science** | Vellore Institute of Technology, India | GPA: 9.0/10 | Jul 2018 – Jun 2022